

# JEE MAIN REVISION MASTER: MOTION IN A STRAIGHT LINE

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 11 Analyzed PYQs

## 1. High-Yield Concept Priority Matrix

| Priority Tier | Concept Name                                                 | Frequency | Category         |
|---------------|--------------------------------------------------------------|-----------|------------------|
| Moderate      | Uniformly Accelerated Rectilinear Motion & Stopping Distance | 6 PYQs    | 1D Kinematics    |
| Moderate      | Motion Under Gravity & Kinematic Graphs                      | 5 PYQs    | Gravity & Graphs |

## 2. Master Formula Reference Card

| Concept / Topic                                              | Mathematical Formulation & Governing Equations                                                                         |
|--------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| Uniformly Accelerated Rectilinear Motion & Stopping Distance | $v^2 = u^2 + 2as$ , $S_n = u + \frac{a}{2}(2n-1)$ , $d_{\text{stop}} = \frac{u^2}{2a}$                                 |
| Motion Under Gravity & Kinematic Graphs                      | $H_{\text{max}} = \frac{u^2}{2g}$ , $T_{\text{total}} = \frac{2u}{g}$ , $\text{Area under } v-t = \text{Displacement}$ |

## 3. Deep-Dive Concepts & Linked PYQ Evidence

### 3.1 Uniformly Accelerated Rectilinear Motion & Stopping Distance (Moderate Yield — 6 PYQs)

**Core Principle:** Equations of motion for constant acceleration. Distance covered in  $n$ -th second  $S_n = u + \frac{a}{2}(2n-1)$ . Stopping distance  $d_s = \frac{u^2}{2a}$  upto  $u^2$ .

■ **Caution / JEE Pitfall:** Stopping distance is proportional to  $u^2$ , NOT  $u$ . Doubling the vehicle speed quadruples the stopping distance!

■ **Speed Trick:** For a body starting from rest with uniform acceleration, distances travelled in successive equal time intervals are in the ratio  $1 : 3 : 5 : 7 : \dots$  (Galileo's odd number ratio).

**Representative Source PYQs:**

- **[2026 Shift] Q#2 (MCQ):** "A car covers the first half of the distance between two places at 40 km/h and another half at 50 km/h. The average speed of the car is..." **[Ans: B]**
- **[2025 Shift] Q#3 (MCQ):** "Two stones begin to fall from rest from the same height, with the second stone starting to fall  $\Delta t$  seconds after the first falls..." **[Ans: N/A]**
- **[2023 Shift] Q#4 (MCQ):** "A body is moving along a straight line with initial velocity  $v_0$ . Its acceleration  $a$  is constant. After  $t$  seconds, its velocity becomes..." **[Ans: N/A]**

### 3.2 Motion Under Gravity & Kinematic Graphs (Moderate Yield — 5 PYQs)

**Core Principle:** Vertical motion under gravity: Maximum height  $H = (u^2 / 2g)$ , time of flight  $T = (2u / g)$ , striking speed  $v = \sqrt{2gh}$ . Slope of  $x$ - $t$  curve gives velocity; slope of  $v$ - $t$  curve gives acceleration; area under  $v$ - $t$  curve gives displacement.

■ **Caution / JEE Pitfall:** At maximum height, vertical velocity is momentarily zero, but acceleration is still  $g = 9.8 \text{ m/s}^2$  downwards!

■ **Speed Trick:** If a ball is dropped from height  $h$  and another is projected vertically downwards with speed  $u$ , the difference in their velocities remains constant.

**Representative Source PYQs:**

- **[2026 Shift] Q#1 (MCQ):** "The velocity of a particle moving along  $x$ -axis is given as  $V=x^2-5x+4$  (in m/s) where  $x$  denotes the  $x$ -coordinate of the particle ..." **[Ans: C]**
- **[2022 Shift] Q#1 (MCQ):** "The displacement  $x$  (in m) of a particle of mass  $m$  (in kg) moving in one dimension under the action of a force, is related to time ..." **[Ans: N/A]**
- **[2021 Shift] Q#3 (MCQ):** "A particle starts from rest. Its acceleration  $a$  versus time  $t$  is shown in the figure. The maximum speed of the particle will be..." **[Ans: N/A]**

## 4. Benchmark Illustrative PYQs with Master Solutions

### Benchmark Problem 1: KCET 2026 (Q#1)

The velocity of a particle moving along  $x$ -axis is given as  $V=x^2-5x+4$  (in m/s) where  $x$  denotes the  $x$ -coordinate of the particle in metres. The magnitude of the acceleration of the particle when the velocity of the particle zero is

- (A)  $2\text{m/s}^2$
- (B)  $3\text{m/s}^2$
- (C) Zero **[Correct]**
- (D)  $1\text{m/s}^2$

**Correct Answer: C**

**Solution Analysis:** The velocity is given as a function of position:  $v=x^2-5x+4$  We need acceleration when the velocity is zero. First find the positions where  $v=0$  :  $x^2-5x+4=0$  Factorising,  $(x-1)(x-4)=0$  So,  $x=1\text{m}$  or  $x=4\text{m}$  Now, since velocity is given as a function of  $x$ , acceleration is not simply  $(dv / dx)$ . We use  $a=(dv / dt)$  Using chain rule,  $a=(dv / dx) \cdot (dx / dt)$  But  $(dx / dt)=v$  So,  $a=v(dv / dx)$  Here, when the velocity is zero,  $v=0$  Hence,  $a=0 \cdot (dv / dx)=0$  So the magnitude of acceleration is  $0\text{m/s}^2$  Correct option: C

**Benchmark Problem 2: KCET 2026 (Q#2)**

A car covers the first half of the distance between two places at 40 km/h and another half at 50 km/h. The average speed of the car is

- (A) 45.00 km/h
- (B) 44.44 km/h **[Correct]**
- (C) 43.14 km/h
- (D) 42.04 km/h

**Correct Answer: B**

**Solution Analysis:** For average speed, we use  $\text{Average speed} = (\text{Total distance} / \text{Total time})$ . Here, the car covers equal distances at two different speeds. Let the total distance be  $2d$  km. So, first half distance  $= d$  km and second half distance  $= d$  km. Time taken for first half:  $(d / 40)$  Time taken for second half:  $(d / 50)$  Total time:  $(d / 40) + (d / 50)$  Now average speed:  $\text{Average speed} = (2d / ((d / 40) + (d / 50)))$  Taking  $d$  common in the denominator:  $\text{Average speed} = (2d / d((1 / 40) + (1 / 50)))$  Cancel  $d$  :  $\text{Average speed} =$